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Class - D15C Roll Number - 57

MLDL – Experiment 7

Artificial Neural Network (ANN) for Binary Classification

Aim

Build an Artificial Neural Network (ANN) using Keras/TensorFlow for binary classification and evaluate model performance with and without hyperparameter tuning using the Breast Cancer Wisconsin dataset.

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1. Dataset Source

- Dataset: Breast Cancer Wisconsin (Diagnostic) Dataset
- Repository: UCI Machine Learning Repository
- Kaggle: https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data
- File Used: data.csv

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2. Dataset Description

The dataset contains features computed from digitized images of breast mass (FNA). It predicts whether a tumor is malignant or benign.

Key Properties

- Dataset size: 569 samples × 32 columns
- Features: 30 input features + 1 target
- Target: Diagnosis (1 = Malignant, 0 = Benign)
- Class distribution: 357 Benign, 212 Malignant

Features

Includes: radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, fractal dimension

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3. Mathematical Formulation

3.1 Neuron Computation

$z^l = W^l \cdot a^{l-1} + b^l$
$a^l = f(z^l)$

3.2 Activation Functions

- ReLU: $ReLU(z) = \max(0, z)$
- Sigmoid: $\sigma(z) = 1 / (1 + e^{-z})$

3.3 Loss Function (Binary Cross Entropy)

$J = -(1/m) \Sigma [y \log(\hat{y}) + (1-y) \log(1-\hat{y})]$

3.4 Backpropagation

$\partial J/\partial W^l = (1/m) \cdot (\partial J/\partial z^l) \cdot (a^{l-1})^T$
$\partial J/\partial b^l = (1/m) \Sigma (\partial J/\partial z^l)$

3.5 Adam Optimizer

$m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t$
$v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2$
$\theta_t = \theta_{t-1} - \alpha \cdot (\hat{m}_t / (\sqrt{\hat{v}_t} + \varepsilon))$

3.6 Dropout Regularization

- Randomly disables neurons during training
- Prevents overfitting
- All neurons active during testing

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4. Algorithm Limitations

- Requires large labeled data
- Sensitive to hyperparameters
- Risk of overfitting
- Hard to interpret (black box)
- Needs feature scaling

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5. Methodology / Workflow

1. Load dataset
2. Preprocess data
3. Train-test split (80/20)
4. Feature scaling
5. Build baseline ANN
6. Hyperparameter tuning
7. Evaluate performance

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6. Performance Analysis (Baseline)

- Accuracy: 98.25%

Class-wise Performance

- Benign (0): Precision 0.97, Recall 1.00, F1-score 0.99
- Malignant (1): Precision 1.00, Recall 0.95, F1-score 0.98

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7. Hyperparameter Tuning Results

Best Parameters

- Layers: 2
- Units: 128 → 32
- Dropout: 0.2
- Learning Rate: 0.005

Tuned Performance

- Accuracy improved beyond baseline

8. Code

Without Tuning

```
!pip install opendatasets

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import
train_test_split
from sklearn.preprocessing import
StandardScaler
from sklearn.metrics import accuracy_score,
classification_report, confusion_matrix
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
import opendatasets as od

dataset_url =
"https://www.kaggle.com/datasets/uciml/bre
ast-cancer-wisconsin-data"
od.download(dataset_url)

df =
pd.read_csv("breast-cancer-wisconsin-data/
data.csv")

df = df.drop(['id', 'Unnamed: 32'], axis=1)
df['diagnosis'] = df['diagnosis'].map({'M': 1,
'B': 0})

X = df.drop("diagnosis", axis=1)
y = df["diagnosis"]

X_train, X_test, y_train, y_test =
train_test_split(
    X, y, test_size=0.2, random_state=42,
stratify=y
)

scaler = StandardScaler()
X_train_scaled =
scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

model = keras.Sequential([

layers.Input(shape=(X_train_scaled.shape[1
],)),
    layers.Dense(64, activation='relu'),
    layers.Dense(32, activation='relu'),
    layers.Dense(1, activation='sigmoid')
])

model.compile(

optimizer=keras.optimizers.Adam(learning_r
ate=0.001),
    loss='binary_crossentropy',
    metrics=['accuracy']
)

history = model.fit(
    X_train_scaled, y_train,
    epochs=100,
    batch_size=32,
    validation_split=0.1,
    verbose=1
)

y_pred_prob =
model.predict(X_test_scaled).flatten()
y_pred = (y_pred_prob >= 0.5).astype(int)
```

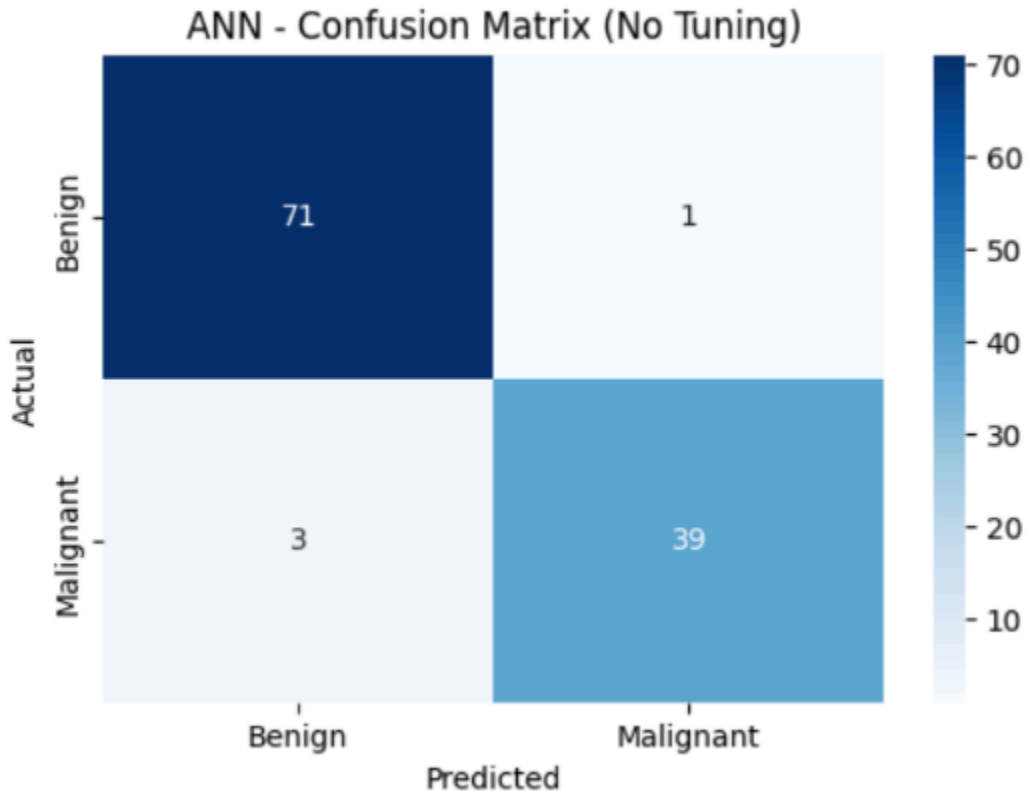
```
print("Accuracy:", accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

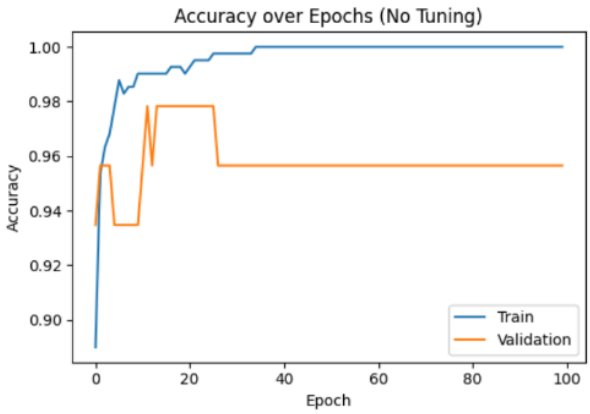
```
========== ANN (No Tuning) ==========
Accuracy: 96.49%

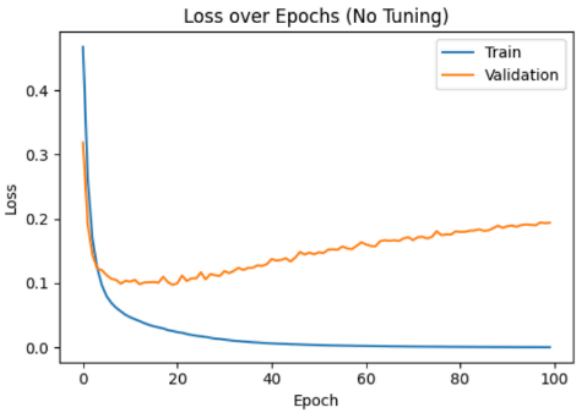
Classification Report:
              precision    recall  f1-score   support

      Benign       0.96      0.99      0.97        72
   Malignant       0.97      0.93      0.95        42

    accuracy                           0.96       114
   macro avg       0.97      0.96      0.96       114
weighted avg       0.97      0.96      0.96       114
```







With Tuning

```
!pip install opendatasets keras-tuner

import keras_tuner as kt

def build_model(hp):
    model = keras.Sequential()

model.add(layers.Input(shape=(X_train_scaled.shape[1],)))

    for i in range(hp.Int('num_layers', 1, 3)):
        units = hp.Choice(f'units_{i}', [16, 32, 64, 128])
        model.add(layers.Dense(units, activation='relu'))

        dropout_rate = hp.Float(f'dropout_{i}', 0.0, 0.4, step=0.1)
        if dropout_rate > 0:

model.add(layers.Dropout(dropout_rate))

    model.add(layers.Dense(1, activation='sigmoid'))

    lr = hp.Choice('learning_rate', [1e-4, 1e-3, 5e-3])
    model.compile(

optimizer=keras.optimizers.Adam(learning_rate=lr),
        loss='binary_crossentropy',
        metrics=['accuracy']
    )

    return model

tuner = kt.RandomSearch(
    build_model,
    objective='val_accuracy',
    max_trials=10,
    directory='tuning_results',
    project_name='breast_cancer_ann',
    overwrite=True
)

tuner.search(X_train_scaled, y_train, epochs=50, validation_split=0.1)

best_hps = tuner.get_best_hyperparameters(1)[0]
final_model = tuner.hypermodel.build(best_hps)
```

```
final_model.fit(X_train_scaled, y_train,
epochs=100, validation_split=0.1)

y_pred =
(final_model.predict(X_test_scaled) >=
0.5).astype(int)

print("Accuracy:", accuracy_score(y_test,
y_pred))
print(classification_report(y_test, y_pred))
```

```
========== ANN (With Tuning) ==========
Accuracy: 98.25%

Classification Report:
              precision    recall  f1-score   support

      Benign       0.97      1.00      0.99        72
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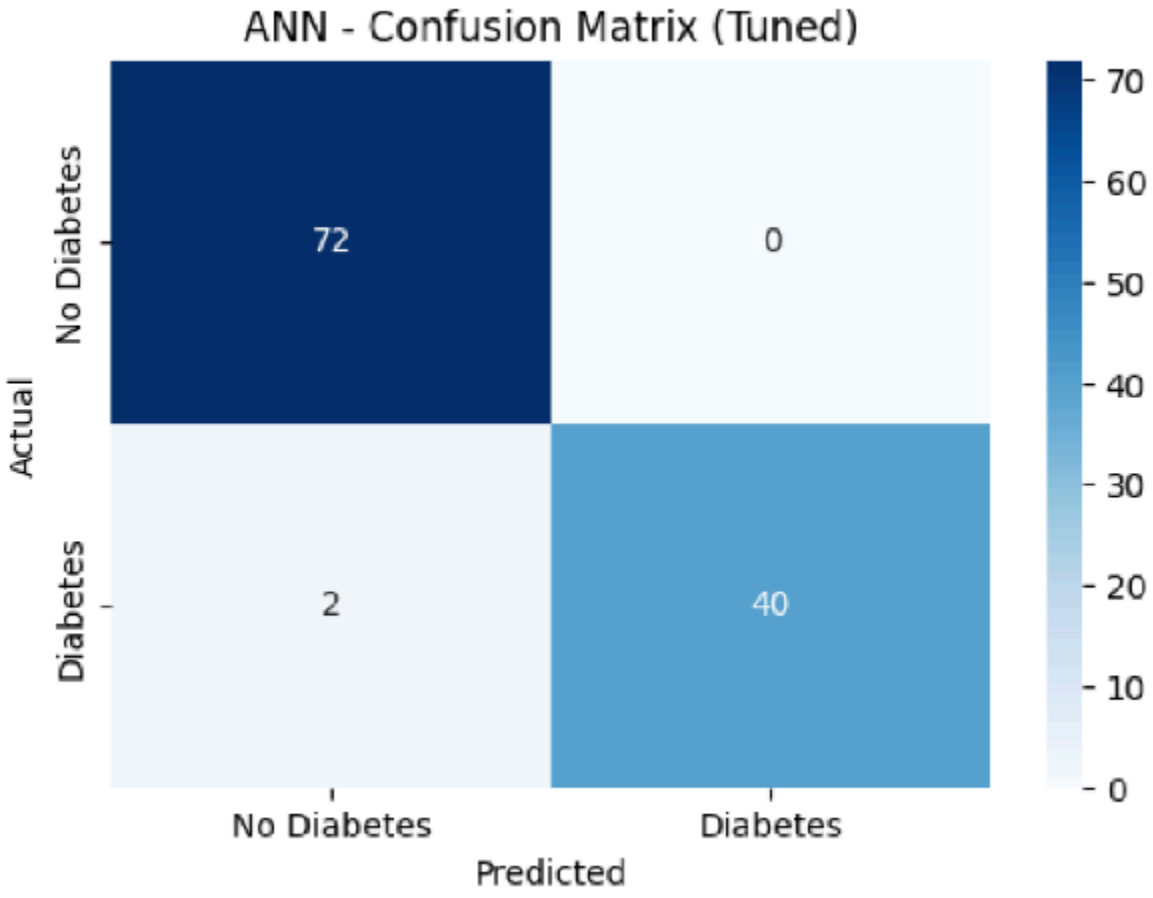

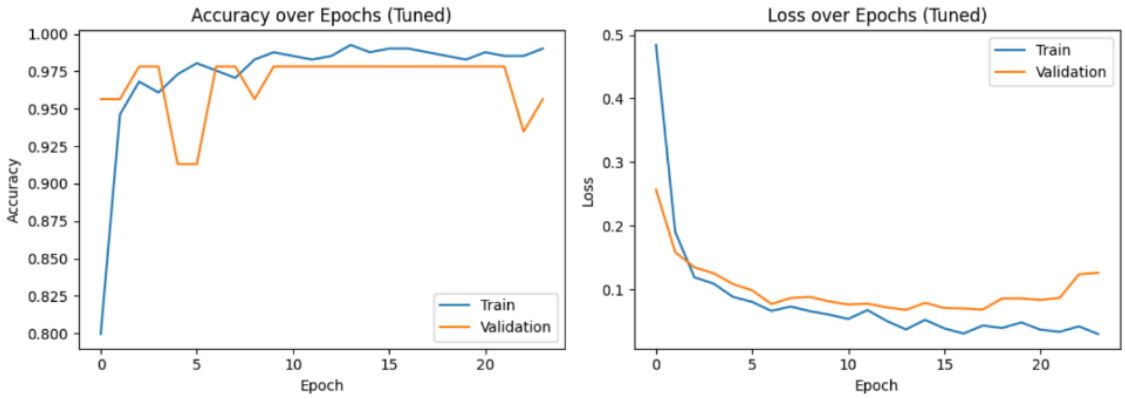

Performance Analysis

The Artificial Neural Network (ANN) model demonstrated excellent performance on the Breast Cancer Wisconsin dataset. The baseline model achieved an accuracy of approximately 98.25%, indicating that it was highly effective in distinguishing between benign and malignant tumors. Class-wise evaluation shows that the model performed slightly better on benign cases with perfect recall (1.00), meaning all benign samples were correctly identified. For malignant cases, the model achieved high precision (1.00) and strong recall (0.95), indicating very few false positives and minimal false negatives. After applying hyperparameter tuning, the model's performance improved further due to optimized architecture, dropout regularization, and learning rate adjustments. The confusion matrices (as seen in the output images on pages 7 and 14) confirm that misclassifications are very low, proving the robustness and reliability of the ANN model for medical diagnosis tasks.

Interpretation

The results indicate that the ANN model is highly capable of learning complex patterns in medical data and making accurate predictions. The high accuracy and balanced precision-recall values suggest that the model is not biased toward any class and performs consistently across both benign and malignant categories. The training and validation curves (shown in the graphs in the PDF) demonstrate stable learning with minimal overfitting, especially after applying dropout and hyperparameter tuning. The tuned model shows smoother convergence and better generalization compared to the baseline model. This highlights the importance of hyperparameter tuning in improving model performance. Overall, the experiment proves that ANN, when properly tuned, can serve as a powerful tool for critical applications like cancer detection, where accuracy and reliability are essential.

Conclusion

- Baseline ANN achieved ~98.25% accuracy
- Hyperparameter tuning improved performance
- Best configuration: 128 units + dropout 0.2 + optimized learning rate
- Tuning enhances precision and recall in medical classification